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(54) **WEARABLE SYSTEM SPEECH PROCESSING**

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Publication Classification

(51) **Int. Cl.**

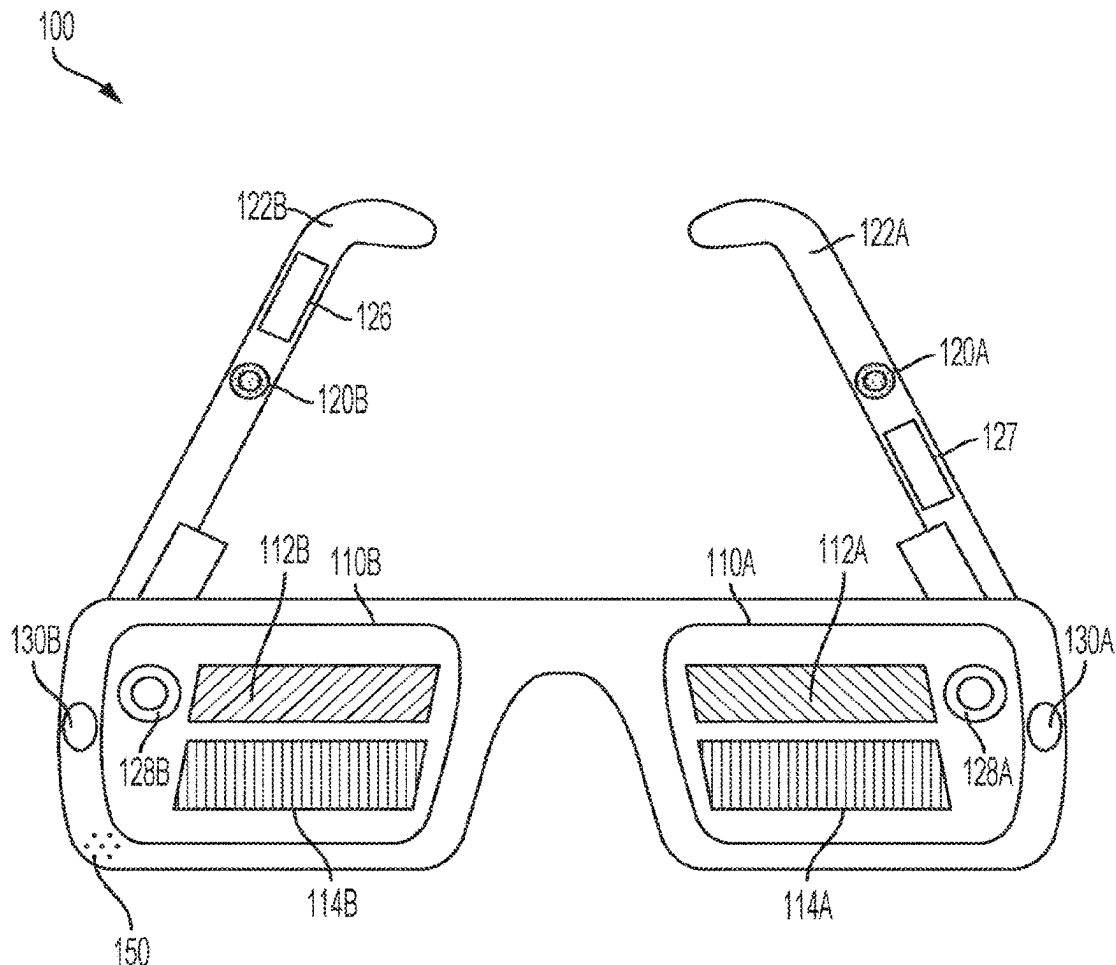
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(57)

ABSTRACT

A method of processing an acoustic signal is disclosed. According to one or more embodiments, a first acoustic signal is received via a first microphone. The first acoustic signal is associated with a first speech of a user of a wearable headgear unit. A first sensor input is received via a sensor. a control parameter is determined based on the sensor input. The control parameter is applied to one or more of the first acoustic signal, the wearable headgear unit, and the first microphone. Determining the control parameter comprises determining, based on the first sensor input, a relationship between the first speech and the first acoustic signal.



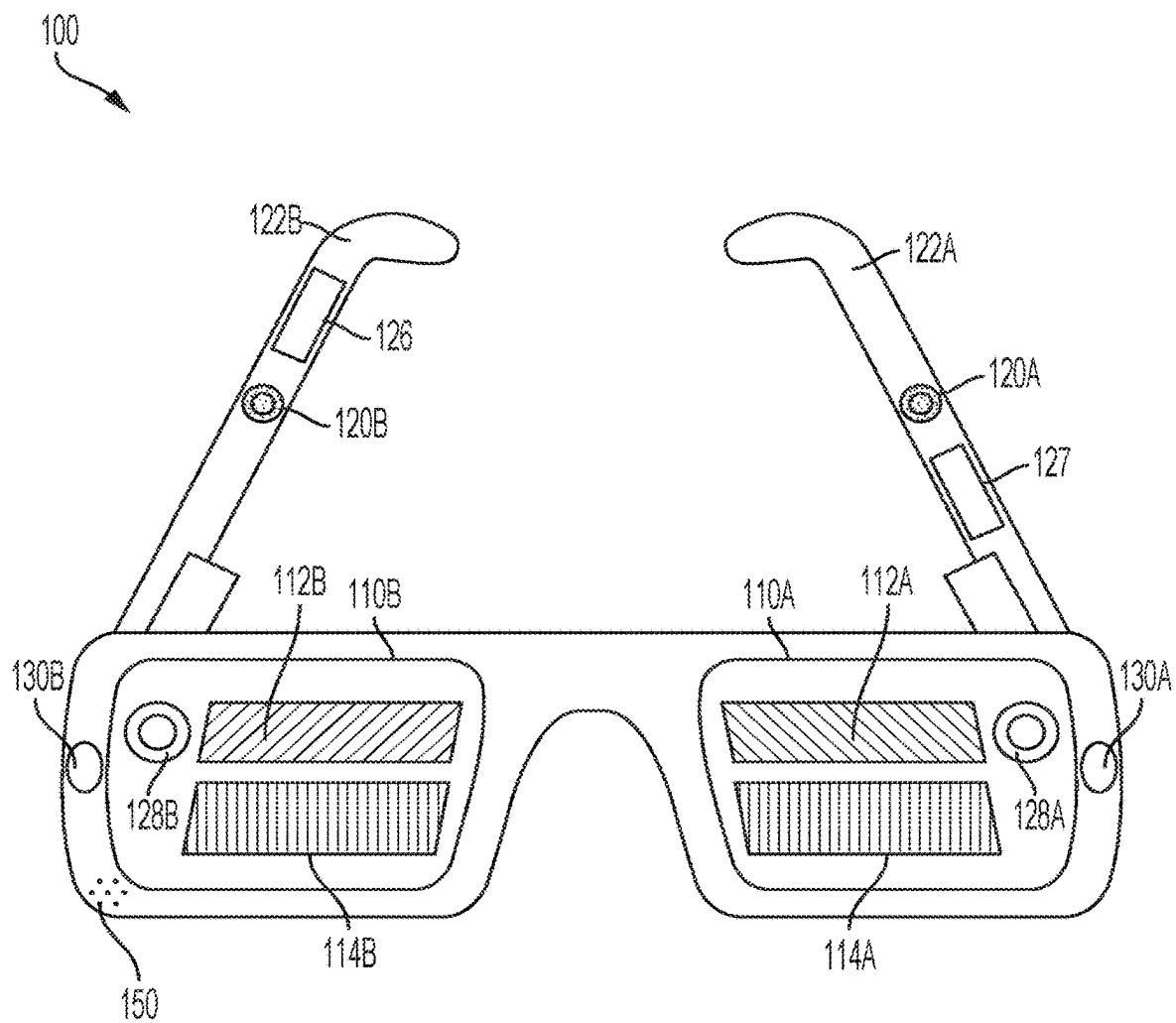


FIG. 1

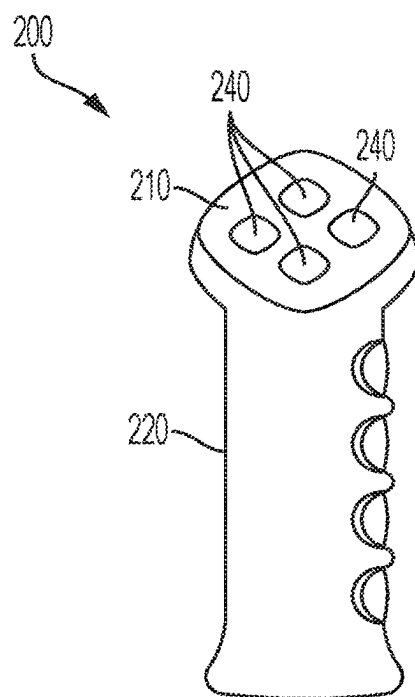


FIG. 2

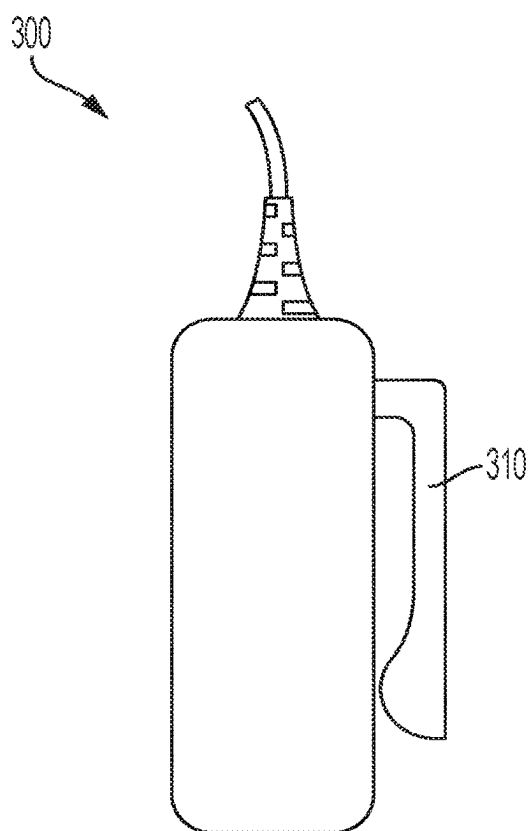


FIG. 3

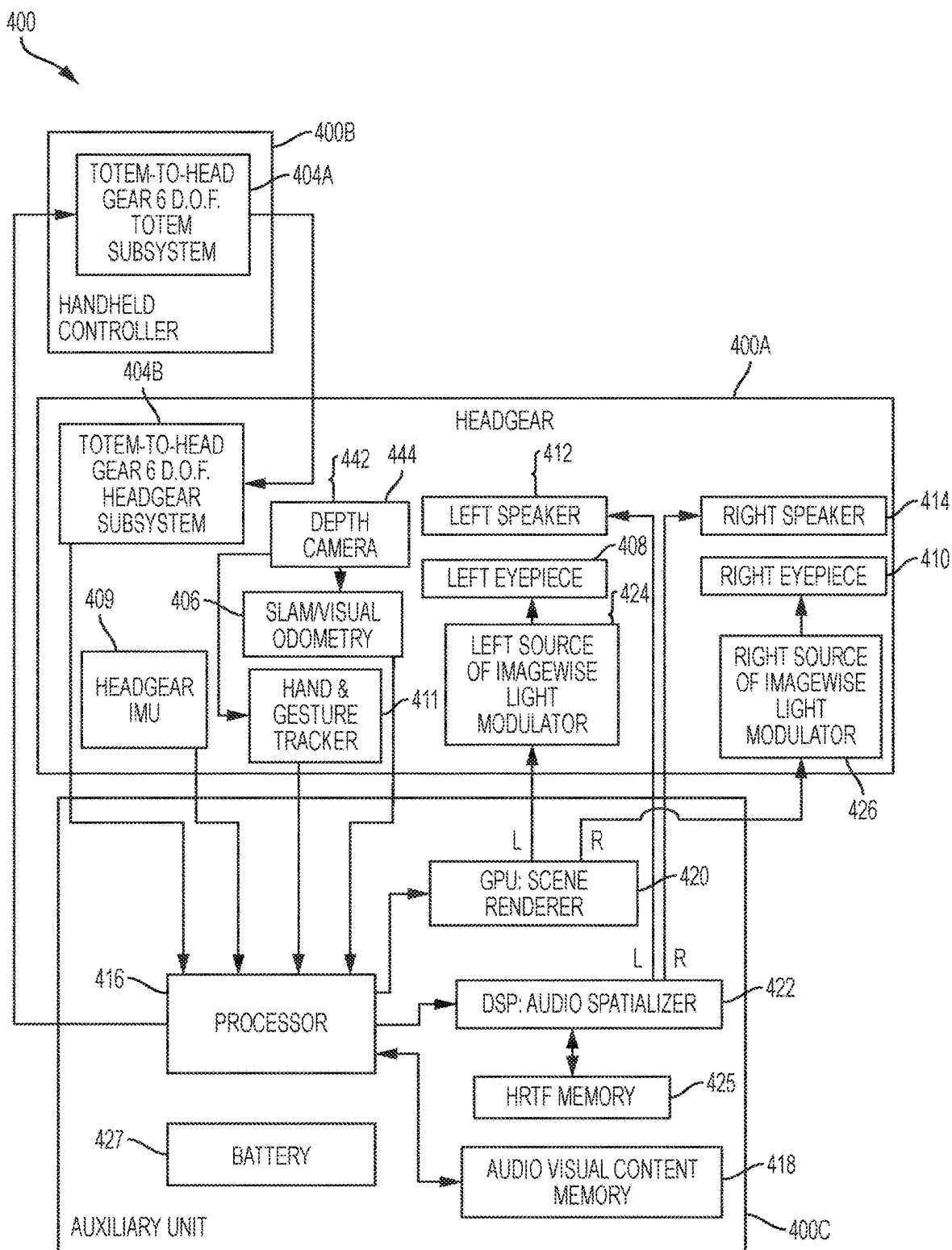


FIG. 4

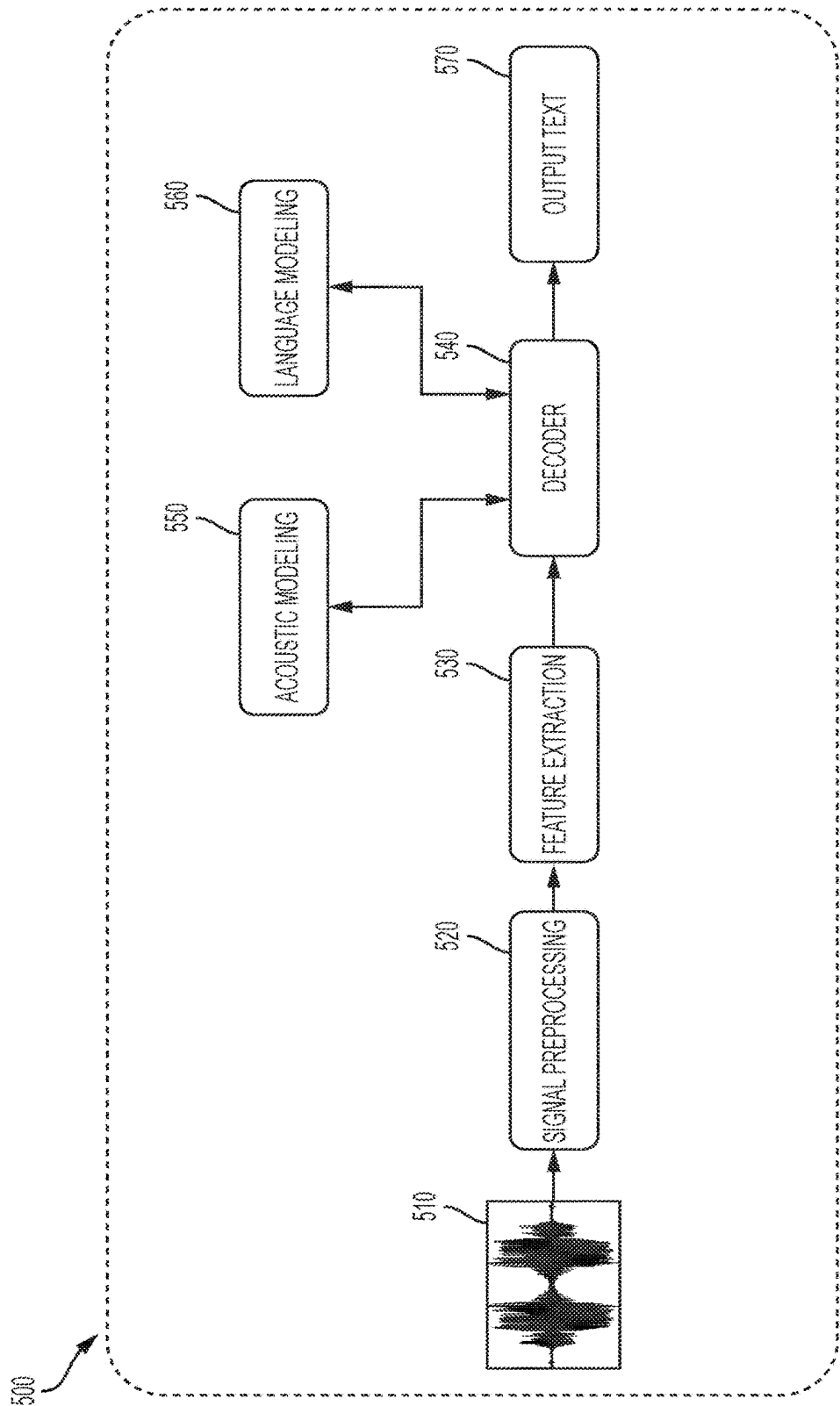


FIG. 5

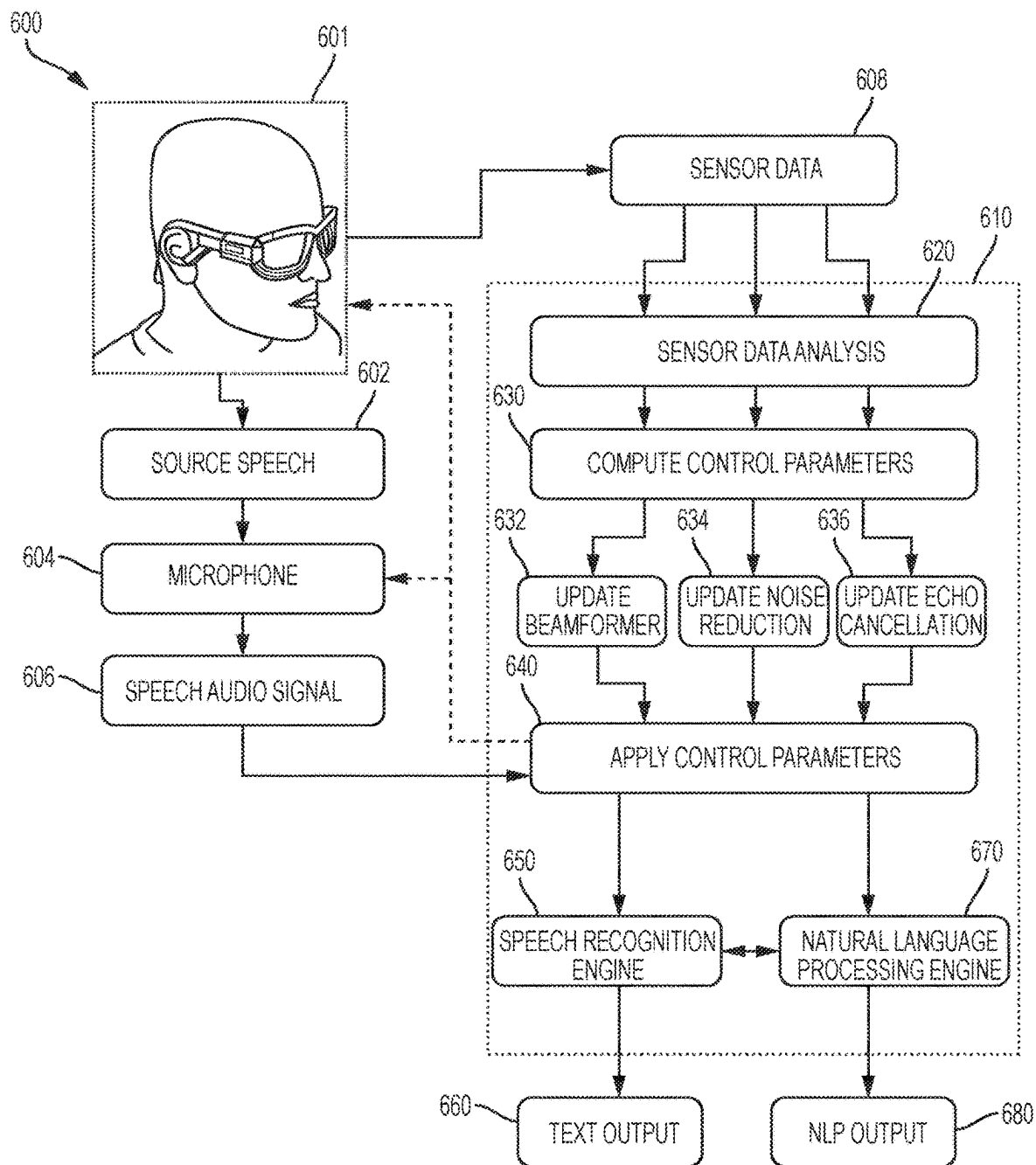


FIG. 6

WEARABLE SYSTEM SPEECH PROCESSING

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 18/510,376, filed Nov. 15, 2023, which is a continuation of U.S. application Ser. No. 17/254,832, filed Dec. 21, 2020, now U.S. Pat. No. 11,854,566, issued Dec. 26, 2023, which is national stage application under 35 U.S.C. § 371 of International Application No. PCT/US2019/038546, filed internationally on Jun. 21, 2019, which claims benefit of U.S. Provisional Patent Application No. 62/687,987, filed Jun. 21, 2018, which are hereby incorporated by reference in their entirety.

FIELD

[0002] This disclosure relates in general to systems and methods for processing acoustic speech signals, and in particular to systems and methods for processing acoustic speech signals generated by a user of a wearable device.

BACKGROUND

[0003] Tools for speech processing are tasked with receiving audio input representing human speech, typically via a microphone, and processing the audio input to determine words, logical structures, or other outputs corresponding to that audio input. For example, automatic speech recognition (ASR) tools may generate a text output based on the human speech corresponding to an audio input signal; and natural language processing (NLP) tools may generate logical structures, or computer data, corresponding to the meaning of that human speech. It is desirable for such processes to happen accurately and quickly—with some applications demanding results in real-time.

[0004] Computer speech processing systems have a history of producing dubious results. In general, the accuracy of a speech processing system can be highly dependent on the quality of the input audio signal, with the highest accuracy obtained from inputs provided under controlled conditions. For example, speech processing systems may perform admirably when the audio input is clearly enunciated speech captured by a microphone at a direct angle, and at close range, with no environmental noise, a high signal/noise ratio, and a constant volume level. However, speech processing systems may struggle to accommodate the many variables that can be introduced into input audio signals by real-world conditions. For example, speech processing signals may demonstrate limited accuracy when multiple speech sources are present (e.g., multiple people speaking at once in the same space); when environmental noise (e.g., wind, rain, electrical interference, ambient noise) blends with a source speech signal; when the human speaker does not enunciate clearly, or speaks with an idiosyncratic or inconsistent tone, accent, or inflection; when the speaker moves or rotates relative to a microphone; when the speaker is in an acoustically reflective environment (e.g., a tiled bathroom or a large cathedral); when the speaker is a large distance from a microphone; when the speaker faces away from the microphone; or when any number of other variables are present and compromise the fidelity of the input audio signal. These problems may be magnified in mobile or outdoor applications, in which unpredictable sources of

noise may be present; and in which attempts to control or understand the speaker's vicinity may be made difficult or impossible.

[0005] It is desirable to use sensor-equipped wearable systems, such as those that incorporate a head-mounted unit, to correct for the effects of such variables on input audio signals for speech processing systems. By presenting speech processing systems with more predictable, higher fidelity input, the outputs of those systems can generate more accurate and more reliable results. In addition, wearable systems are well suited for mobile, outdoor applications—precisely the type of applications in which many conventional speech processing systems may perform especially poorly.

BRIEF SUMMARY

[0006] Examples of the disclosure describe systems and methods for processing acoustic signals. According to one or more embodiments, a first acoustic signal is received via a first microphone. The first acoustic signal is associated with a first speech of a user of a wearable headgear unit. A first sensor input is received via a sensor. A control parameter is determined based on the sensor input. The control parameter is applied to one or more of the first acoustic signal, the wearable headgear unit, and the first microphone. Determining the control parameter comprises determining, based on the first sensor input, a relationship between the first speech and the first acoustic signal.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 illustrates an example wearable head device that can be used as part of a wearable system, according to some embodiments.

[0008] FIG. 2 illustrates an example handheld controller that can be used as part of a wearable system, according to some embodiments.

[0009] FIG. 3 illustrates an example auxiliary unit that can be used as part of a wearable system, according to some embodiments.

[0010] FIG. 4 illustrates an example functional block diagram for an example wearable system, according to some embodiments.

[0011] FIG. 5 illustrates a flow chart of an example speech processing system, according to some embodiments.

[0012] FIG. 6 illustrates a flow chart of an example system for processing acoustic speech signals, according to some embodiments.

DETAILED DESCRIPTION

[0013] In the following description of examples, reference is made to the accompanying drawings which form a part hereof, and in which it is shown by way of illustration specific examples that can be practiced. It is to be understood that other examples can be used and structural changes can be made without departing from the scope of the disclosed examples.

Example Wearable System

[0014] FIG. 1 illustrates an example wearable head device 100 configured to be worn on the head of a user. Wearable head device 100 may be part of a broader wearable system that comprises one or more components, such as a head device (e.g., wearable head device 100), a handheld con-

troller (e.g., handheld controller **200** described below), and/or an auxiliary unit (e.g., auxiliary unit **300** described below). In some examples, wearable head device **100** can be used for virtual reality, augmented reality, or mixed reality systems or applications. Wearable head device **100** can comprise one or more displays, such as displays **110A** and **110B** (which may comprise left and right transmissive displays, and associated components for coupling light from the displays to the user's eyes, such as orthogonal pupil expansion (OPE) grating sets **112A/112B** and exit pupil expansion (EPE) grating sets **114A/114B**); left and right acoustic structures, such as speakers **120A** and **120B** (which may be mounted on temple arms **122A** and **122B**, and positioned adjacent to the user's left and right ears, respectively); one or more sensors such as infrared sensors, accelerometers, GPS units, inertial measurement units (IMU) (e.g., IMU **126**), acoustic sensors (e.g., microphone **150**); orthogonal coil electromagnetic receivers (e.g., receiver **127** shown mounted to the left temple arm **122A**); left and right cameras (e.g., depth (time-of-flight) cameras **130A** and **130B**) oriented away from the user; and left and right eye cameras oriented toward the user (e.g., for detecting the user's eye movements) (e.g., eye cameras **128** and **128B**). However, wearable head device **100** can incorporate any suitable display technology, and any suitable number, type, or combination of sensors or other components without departing from the scope of the invention. In some examples, wearable head device **100** may incorporate one or more microphones **150** configured to detect audio signals generated by the user's voice; such microphones may be positioned in a wearable head device adjacent to the user's mouth. In some examples, wearable head device **100** may incorporate networking features (e.g., Wi-Fi capability) to communicate with other devices and systems, including other wearable systems. Wearable head device **100** may further include components such as a battery, a processor, a memory, a storage unit, or various input devices (e.g., buttons, touchpads); or may be coupled to a handheld controller (e.g., handheld controller **200**) or an auxiliary unit (e.g., auxiliary unit **300**) that comprises one or more such components. In some examples, sensors may be configured to output a set of coordinates of the head-mounted unit relative to the user's environment, and may provide input to a processor performing a Simultaneous Localization and Mapping (SLAM) procedure and/or a visual odometry algorithm. In some examples, wearable head device **100** may be coupled to a handheld controller **200**, and/or an auxiliary unit **300**, as described further below.

[0015] FIG. 2 illustrates an example mobile handheld controller component **200** of an example wearable system. In some examples, handheld controller **200** may be in wired or wireless communication with wearable head device **100** and/or auxiliary unit **300** described below. In some examples, handheld controller **200** includes a handle portion **220** to be held by a user, and one or more buttons **240** disposed along a top surface **210**. In some examples, handheld controller **200** may be configured for use as an optical tracking target; for example, a sensor (e.g., a camera or other optical sensor) of wearable head device **100** can be configured to detect a position and/or orientation of handheld controller **200**—which may, by extension, indicate a position and/or orientation of the hand of a user holding handheld controller **200**. In some examples, handheld controller **200** may include a processor, a memory, a storage unit, a

display, or one or more input devices, such as described above. In some examples, handheld controller **200** includes one or more sensors (e.g., any of the sensors or tracking components described above with respect to wearable head device **100**). In some examples, sensors can detect a position or orientation of handheld controller **200** relative to wearable head device **100** or to another component of a wearable system. In some examples, sensors may be positioned in handle portion **220** of handheld controller **200**, and/or may be mechanically coupled to the handheld controller. Handheld controller **200** can be configured to provide one or more output signals, corresponding, for example, to a pressed state of the buttons **240**; or a position, orientation, and/or motion of the handheld controller **200** (e.g., via an IMU). Such output signals may be used as input to a processor of wearable head device **100**, to auxiliary unit **300**, or to another component of a wearable system. In some examples, handheld controller **200** can include one or more microphones to detect sounds (e.g., a user's speech, environmental sounds), and in some cases provide a signal corresponding to the detected sound to a processor (e.g., a processor of wearable head device **100**).

[0016] FIG. 3 illustrates an example auxiliary unit **300** of an example wearable system. In some examples, auxiliary unit **300** may be in wired or wireless communication with wearable head device **100** and/or handheld controller **200**. The auxiliary unit **300** can include a battery to provide energy to operate one or more components of a wearable system, such as wearable head device **100** and/or handheld controller **200** (including displays, sensors, acoustic structures, processors, microphones, and/or other components of wearable head device **100** or handheld controller **200**). In some examples, auxiliary unit **300** may include a processor, a memory, a storage unit, a display, one or more input devices, and/or one or more sensors, such as described above. In some examples, auxiliary unit **300** includes a clip **310** for attaching the auxiliary unit to a user (e.g., a belt worn by the user). An advantage of using auxiliary unit **300** to house one or more components of a wearable system is that doing so may allow large or heavy components to be carried on a user's waist, chest, or back—which are relatively well suited to support large and heavy objects—rather than mounted to the user's head (e.g., if housed in wearable head device **100**) or carried by the user's hand (e.g., if housed in handheld controller **200**). This may be particularly advantageous for relatively heavy or bulky components, such as batteries.

[0017] FIG. 4 shows an example functional block diagram that may correspond to an example wearable system **400**, such as may include example wearable head device **100**, handheld controller **200**, and auxiliary unit **300** described above. In some examples, the wearable system **400** could be used for virtual reality, augmented reality, or mixed reality applications. As shown in FIG. 4, wearable system **400** can include example handheld controller **400B**, referred to here as a “totem” (and which may correspond to handheld controller **200** described above); the handheld controller **400B** can include a totem-to-headgear six degree of freedom (6DOF) totem subsystem **404A**. Wearable system **400** can also include example wearable head device **400A** (which may correspond to wearable headgear device **100** described above); the wearable head device **400A** includes a totem-to-headgear 6DOF headgear subsystem **404B**. In the example, the 6DOF totem subsystem **404A** and the 6DOF

headgear subsystem **404B** cooperate to determine six coordinates (e.g., offsets in three translation directions and rotation along three axes) of the handheld controller **400B** relative to the wearable head device **400A**. The six degrees of freedom may be expressed relative to a coordinate system of the wearable head device **400A**. The three translation offsets may be expressed as X, Y, and Z offsets in such a coordinate system, as a translation matrix, or as some other representation. The rotation degrees of freedom may be expressed as sequence of yaw, pitch and roll rotations; as vectors; as a rotation matrix; as a quaternion; or as some other representation. In some examples, one or more depth cameras **444** (and/or one or more non-depth cameras) included in the wearable head device **400A**; and/or one or more optical targets (e.g., buttons **240** of handheld controller **200** as described above, or dedicated optical targets included in the handheld controller) can be used for 6DOF tracking. In some examples, the handheld controller **400B** can include a camera, as described above; and the headgear **400A** can include an optical target for optical tracking in conjunction with the camera. In some examples, the wearable head device **400A** and the handheld controller **400B** each include a set of three orthogonally oriented solenoids which are used to wirelessly send and receive three distinguishable signals. By measuring the relative magnitude of the three distinguishable signals received in each of the coils used for receiving, the 6DOF of the handheld controller **400B** relative to the wearable head device **400A** may be determined. In some examples, 6DOF totem subsystem **404A** can include an Inertial Measurement Unit (IMU) that is useful to provide improved accuracy and/or more timely information on rapid movements of the handheld controller **400B**.

[0018] In some examples involving augmented reality or mixed reality applications, it may be desirable to transform coordinates from a local coordinate space (e.g., a coordinate space fixed relative to wearable head device **400A**) to an inertial coordinate space, or to an environmental coordinate space. For instance, such transformations may be necessary for a display of wearable head device **400A** to present a virtual object at an expected position and orientation relative to the real environment (e.g., a virtual person sitting in a real chair, facing forward, regardless of the position and orientation of wearable head device **400A**), rather than at a fixed position and orientation on the display (e.g., at the same position in the display of wearable head device **400A**). This can maintain an illusion that the virtual object exists in the real environment (and does not, for example, appear positioned unnaturally in the real environment as the wearable head device **400A** shifts and rotates). In some examples, a compensatory transformation between coordinate spaces can be determined by processing imagery from the depth cameras **444** (e.g., using a Simultaneous Localization and Mapping (SLAM) and/or visual odometry procedure) in order to determine the transformation of the wearable head device **400A** relative to an inertial or environmental coordinate system. In the example shown in FIG. 4, the depth cameras **444** can be coupled to a SLAM/visual odometry block **406** and can provide imagery to block **406**. The SLAM/visual odometry block **406** implementation can include a processor configured to process this imagery and determine a position and orientation of the user's head, which can then be used to identify a transformation between a head coordinate space and a real coordinate space. Similarly, in some examples, an additional source of information

on the user's head pose and location is obtained from an IMU **409** of wearable head device **400A**.

[0019] Information from the IMU **409** can be integrated with information from the SLAM/visual odometry block **406** to provide improved accuracy and/or more timely information on rapid adjustments of the user's head pose and position.

[0020] In some examples, the depth cameras **444** can supply 3D imagery to a hand gesture tracker **411**, which may be implemented in a processor of wearable head device **400A**. The hand gesture tracker **411** can identify a user's hand gestures, for example by matching 3D imagery received from the depth cameras **444** to stored patterns representing hand gestures. Other suitable techniques of identifying a user's hand gestures will be apparent.

[0021] In some examples, one or more processors **416** may be configured to receive data from headgear subsystem **404B**, the IMU **409**, the SLAM/visual odometry block **406**, depth cameras **444**, a microphone **450**; and/or the hand gesture tracker **411**. The processor **416** can also send and receive control signals from the 6DOF totem system **404A**. The processor **416** may be coupled to the 6DOF totem system **404A** wirelessly, such as in examples where the handheld controller **400B** is untethered. Processor **416** may further communicate with additional components, such as an audio-visual content memory **418**, a Graphical Processing Unit (GPU) **420**, and/or a Digital Signal Processor (DSP) audio spatializer **422**. The DSP audio spatializer **422** may be coupled to a Head Related Transfer Function (HRTF) memory **425**. The GPU **420** can include a left channel output coupled to the left source of imagewise modulated light **424** and a right channel output coupled to the right source of imagewise modulated light **426**. GPU **420** can output stereoscopic image data to the sources of imagewise modulated light **424**, **426**. The DSP audio spatializer **422** can output audio to a left speaker **412** and/or a right speaker **414**. The DSP audio spatializer **422** can receive input from processor **419** indicating a direction vector from a user to a virtual sound source (which may be moved by the user, e.g., via the handheld controller **400B**). Based on the direction vector, the DSP audio spatializer **422** can determine a corresponding HRTF (e.g., by accessing a HRTF, or by interpolating multiple HRTFs). The DSP audio spatializer **422** can then apply the determined HRTF to an audio signal, such as an audio signal corresponding to a virtual sound generated by a virtual object. This can enhance the believability and realism of the virtual sound, by incorporating the relative position and orientation of the user relative to the virtual sound in the mixed reality environment—that is, by presenting a virtual sound that matches a user's expectations of what that virtual sound would sound like if it were a real sound in a real environment.

[0022] In some examples, such as shown in FIG. 4, one or more of processor **416**, GPU **420**, DSP audio spatializer **422**, HRTF memory **425**, and audio/visual content memory **418** may be included in an auxiliary unit **400C** (which may correspond to auxiliary unit **300** described above). The auxiliary unit **400C** may include a battery **427** to power its components and/or to supply power to wearable head device **400A** and/or handheld controller **400B**. Including such components in an auxiliary unit, which can be mounted to a user's waist, can limit the size and weight of wearable head device **400A**, which can in turn reduce fatigue of a user's head and neck.

[0023] While FIG. 4 presents elements corresponding to various components of an example wearable system 400, various other suitable arrangements of these components will become apparent to those skilled in the art. For example, elements presented in FIG. 4 as being associated with auxiliary unit 400C could instead be associated with wearable head device 400A or handheld controller 400B. Furthermore, some wearable systems may forgo entirely a handheld controller 400B or auxiliary unit 400C. Such changes and modifications are to be understood as being included within the scope of the disclosed examples.

Speech Processing Engines

[0024] Speech processing systems in general include systems that accept an input audio signal corresponding to human speech (a source signal); process and analyze the input audio signal; and produce, as a result of the analysis, an output corresponding to the human speech. A process or module for performing these tasks may be considered a speech processing engine. In the case of automatic speech recognition systems, the output of a speech processing engine may be a text transcription of the human speech. In the case of natural language processing systems, the output may be one or more commands or instructions indicated by the human speech; or some representation (e.g., a logical expression or a data structure) of the semantic meaning of the human speech. Other types of speech processing systems (e.g., automatic translation systems) are contemplated and are within the scope of the disclosure.

[0025] Speech processing systems are found in a diverse array of products and applications: conventional telephone systems; automated voice messaging systems; voice assistants (including standalone and smartphone-based voice assistants); vehicles and aircraft; desktop and document processing software; data entry; home appliances; medical devices; language translation software; closed captioning systems; and others. An advantage of speech processing systems is that they may allow users to provide input to a computer system using natural spoken language, such as presented to a microphone, instead of conventional computer input devices such as keyboards or touch panels; accordingly, speech processing systems may be particularly useful in environments where conventional input devices (e.g., keyboards) may be unavailable or impractical. Further, by permitting users to provide intuitive voice-based input, speech recognition systems can heighten feelings of immersion. As such, speech recognition can be a natural fit for wearable systems, and in particular, for virtual reality, augmented reality, and/or mixed reality applications of wearable systems, in which user immersion is a primary goal, and in which it may be desirable to limit the use of conventional computer input devices, whose presence may detract from feelings of immersion.

[0026] FIG. 5 shows an automatic speech recognition engine 500, according to some embodiments. Engine 500 is intended to be illustrative of automatic speech recognition systems in general; other specific systems are possible and within the scope of the disclosure. Engine 500 may be implemented using one or more processors (e.g., CPUs, GPUs, and/or DSPs), a memory, an input device (e.g., a microphone), an output device (e.g., a display, a speaker), a network, a database, and/or other suitable components. In engine 500, an audio signal 510, corresponding to a source human speech signal, is presented to a signal preprocessing

stage 520. In some examples, signal preprocessing stage 520 can apply one or more signal processing functions to audio signal 510. For example, preprocessing functions can include audio processing functions, such as peak compression; noise reduction; bandlimiting; equalization; signal attenuation; or other suitable functions. These preprocessing functions can simplify the task of later processing and analyzing audio signal 510. For instance, feature extraction algorithms can be calibrated to perform best on input signals having certain audio characteristics, such as gain and frequency characteristics, and where a signal/noise ratio of the input signal is maximized. In some examples, preprocessing audio signal 510 can condition the signal such that the signal can be more reliably analyzed elsewhere in engine 500. For instance, signal preprocessing stage 520 can re-encode audio signal 510 (e.g., re-encoding at a specific bitrate); or convert audio signal 510 from a first form (e.g., a time-domain signal) into a second form (e.g., a frequency-domain signal; or a parametric representation) that may simplify the later processing of audio signal 510. In some examples, one or more functions of preprocessing stage 520 may be performed by a DSP.

[0027] At stage 530, a feature extraction process can be applied to audio signal 510 (as preprocessed at stage 520). A goal of feature extraction is to identify individual speech features of audio signal 510, and reduce or eliminate variations in these features so that the features can be effectively and consistently processed (e.g., compared against stored patterns in a database). For example, feature extraction can reduce, eliminate, or control variations in the speaker's pitch, gender, accent, pronunciation, and pacing. Feature extraction can also reduce, eliminate, or control variations in recording equipment (e.g., microphone type), signal transmission (e.g., via land-based telephone lines or cellular telephone networks), or recording environment (e.g., room acoustics; background noise levels; speaker's distance from microphone; speaker's angle to microphone). Various suitable techniques for feature extraction are known in the art.

[0028] Speech features extracted from audio signal 510 at stage 530 can be presented to a decoder stage 540. The goal of decoder stage 540 is to determine a text output 570 that corresponds to the source human speech from which audio signal 510 was generated. In some examples, text output 570 need not be text, but can be another data representation of the source speech. Various techniques exist for decoding speech features into text; for example, hidden Markov models; Viterbi decoding; beam search; dynamic search; multipass search; weighted finite state transducers (WFST); or any suitable combination of the above. Other suitable techniques will be familiar to those skilled in the art.

[0029] In some examples, decoder 540 can make use of an acoustic modeling stage 550 to facilitate generating text output 570. Acoustic modeling stage 550 can use a model of the relationships between speech signals and linguistic units (e.g., phonemes) to identify one or more linguistic units from an audio signal 510 (including one or more features extracted at stage 530). Various suitable acoustic modeling techniques that can be applied at acoustic modeling stage 550 will be familiar to those skilled in the art.

[0030] In some examples, decoder 540 can make use of a language modeling stage 560 to facilitate generating text output 570. Linguistic modeling stage 560 can use a model of a language's grammar, vocabulary, and other characteristics to determine linguistic units (e.g., phonemes) that most

probably correspond to features of audio signal **510**. For example, a linguistic model applied by stage **560** could conclude that a particular extracted feature is more likely to correspond to a word that appears with a high frequency in the speaker's language than to a word that appears with a low frequency in that language. Various suitable linguistic modeling techniques that can be applied at linguistic modeling stage **560** will be familiar to those skilled in the art. In addition, decoder **540** may make use of other suitable techniques or models to facilitate the generation of text output **570**; the disclosure is not limited to any particular technique or group of techniques.

[0031] Typically, text output **570** does not correspond to the source human speech with perfect certainty; instead, the likelihood that text output **570** correctly corresponds to the source human speech can be expressed as some probability or confidence interval. Because of the many variables that can affect audio signal **510**, even sophisticated speech recognition systems do not consistently produce perfect text output for all speakers. For example, the reliability of speech recognition systems such as engine **500** may be highly dependent on the quality of the input audio signal **510**. Where audio signal **510** is recorded in ideal conditions—for example, in acoustically controlled environments, with the human speaker enunciating clearly and directly into a microphone from a close distance—the source speech can be more readily determined from the audio signal. For example, features can be more reliably extracted from audio signal **510** at stage **530**; and decoder **540** can more effectively determine which text output **570** corresponds to those features (e.g., acoustic modeling can be more reliably applied to the features at stage **550**, and/or linguistic modeling can be more reliably applied to the features at stage **560**).

[0032] In real-world applications, however, audio signal **510** may deviate from ideal conditions, such that determining the source human speech becomes more difficult. For example, audio signal **510** may incorporate environmental noise, such as may be introduced by outdoor environments, or by substantial distances between a human speaker and a microphone; electrical noise, such as from electrical interference (e.g., battery chargers for a smartphone); natural reverberations, such as from nearby surfaces (e.g., concrete, bathroom tile) or acoustic spaces (e.g., caverns, cathedrals); or other unwanted effects. In addition, audio signal **510** may suffer from attenuation of certain frequencies, such as may result when a human speaker faces away from a microphone; this is particularly problematic when the frequencies attenuated are those that carry significant speech-related information (e.g., formant frequencies that can be used to distinguish between vowel sounds). Similarly, audio signal **510** may suffer from overall low amplitude, or low signal/noise ratio, which may be caused when a large distance sits between a human speaker and a microphone. In addition, if the human speaker moves and reorients himself or herself while speaking, the audio signal **510** may change characteristics over the course of the signal, further complicating efforts to determine the underlying speech.

[0033] Although example system **500** illustrates an example speech recognition engine, other types of speech processing engines may follow an analogous structure: for example, a natural language processing engine, upon accepting an input audio signal corresponding to human speech, may perform a signal preprocessing stage; extract components from the signal (e.g., via segmentation and/or tokeni-

zation stages); and perform detection/analysis of the components, in some cases with the assistance of one or more linguistic modeling subsystems. Moreover, in some examples, the output of an automatic speech recognition engine, such as shown in example system **500**, may be used as input to a further language processing engine; for instance, a natural language processing engine could accept as input the text output **570** of example system **500**. Such systems may experience challenges analogous to those faced by example system **500**. For example, variations in an input audio signal that make it more difficult to recover a source speech signal underlying the audio signal, such as described above, may also make it more difficult to provide other forms of output (e.g., logical expressions or data structures in the case of natural language processing engines). Accordingly, such systems also stand to benefit from the present invention, such as described below.

Improving Speech Processing Using Wearable Systems

[0034] The present disclosure is directed to systems and methods for improving the accuracy of speech processing systems by using input from sensors, for example those associated with wearable devices (e.g., head-mounted devices such as described above with respect to FIG. 1) to reduce, eliminate, or control variations in input audio signals, such as those described above with respect to audio signal **510**. Such variations may be especially pronounced in mobile applications of speech processing, or in applications of speech processing in uncontrolled environments, such as outdoor environments. Wearable systems are frequently intended for use in such applications, and may experience such variations. The wearable system may generally refer to any combination of a head device (e.g., wearable head device **100**), a handheld controller (e.g., handheld controller **200**), an auxiliary unit (e.g., auxiliary unit **300**), and/or an environment of the head device. In some embodiments, the sensors of the wearable system may be on the head device, the handheld controller, the auxiliary unit, and/or in the environment of the head device. For example, because wearable systems may be designed to be mobile, audio signals recorded at a single stationary microphone (e.g., at a standalone voice assistant device) from a user of a wearable system may suffer from a low signal/noise ratio if the user is far from the single stationary microphone, or from “acoustic shadows” or unwanted frequency response if the user faces away from the microphone. Moreover, the audio signal may change characteristics over time as the user moves and turns with respect to the single stationary microphone, as may be expected by a mobile user of a wearable system. In addition, because some wearable systems are intended for use in uncontrolled environments, a high potential exists for environmental noise (or speech of other humans) to be recorded along with the target human speech. Similarly, such uncontrolled environments can introduce unwanted echoes and reverberations into audio signal **510** that can obscure the underlying speech.

[0035] As described above with respect to example wearable head device **100** in FIG. 1, a wearable system can include one or more sensors that can provide input about the user and/or the environment of the wearable system. For instance, wearable head device **100** can include a camera (e.g., camera **444** described in FIG. 4) to output visual signals corresponding to the environment; in some examples, the camera can be a forward-facing camera on a

head-mounted unit that shows what is currently in front of the user of the wearable system. In some examples, wearable head device **100** can include a LIDAR unit, a radar unit, and/or acoustic sensors, which can output signals corresponding to the physical geometry (e.g., walls, physical objects) of the user's environment. In some examples, wearable head device **100** can include a GPS unit, which can indicate geographic coordinates corresponding to the wearable system's current location. In some examples, wearable head device **100** can include an accelerometer, a gyroscope; and/or an inertial measurement unit (IMU) to indicate an orientation of the wearable head device **100**. In some examples, wearable head device **100** can include environmental sensors, such as temperature or pressure sensors. In some examples, wearable head device **100** can include biometric sensors, such as iris cameras; fingerprint sensors; eye tracking sensors; or sensors to measure a user's vital signs. In examples where wearable head device **100** includes a head-mounted unit, such orientation can correspond to an orientation of the user's head (and, by extension, the user's mouth and a direction of the user's speech). Other suitable sensors can be included. In some embodiments, the handheld controller **200**, the auxiliary unit **300**, and/or the environment of the wearable head device **100** can include any suitable one or more of the sensors mentioned above with respect to the wearable head device **100**. In addition, in some cases, one or more sensors may be placed in an environment with which the wearable system interacts. For instance, a wearable system may be designed to be worn by a driver of an automobile, and appropriate sensors (e.g., depth cameras, accelerometers, etc.) may be placed in the interior of the automobile. One advantage of this approach is that the sensors can occupy a known location within the environment; compared to sensors that can be attached to a wearable device that moves within the environment, this configuration can simplify the interpretation of data provided by those sensors.

[0036] The signals provided by such sensors of the wearable system (e.g., the wearable head device **100**, the handheld controller **200**, the auxiliary unit **300**, and/or the environment of the wearable head device **100**) can be used to provide information about the characteristics of an audio signal recorded by the wearable system, and/or information about the relationship between the audio signal and an underlying source speech signal. This information can in turn be used to more effectively determine source speech underlying that audio signal.

[0037] To illustrate, FIG. 6 shows an example speech recognition system **600** that incorporates a wearable system to improve speech recognition of an audio signal recorded by one or more microphones. FIG. 6 shows a user of a wearable system **601**, which may correspond to wearable system **400** described above, and which may include one or more of example wearable head device **100**; handheld controller **200**; and auxiliary unit **300**. The user of wearable system **601** provides a verbal utterance **602** (the "source speech") which is detected at one or more microphones **604**, which output corresponding audio signal **606**. Wearable system **601** can include one or more sensors described above, including one or more of a camera; a LIDAR unit; a radar unit; an acoustic sensor; a GPS unit; an accelerometer; a gyroscope; an IMU; a microphone (which may be one of microphones **604**); a temperature sensor; biometric sensors; or any other suitable sensor or combination of sensors.

Wearable system **601** can also include one or more processors (e.g., CPUs, GPUs, and/or DSPs), a memory, an input device (e.g., a microphone), an output device (e.g., a display, a speaker), a network, and/or a database. These components may be implemented using a combination of a wearable head device **100**, a handheld controller **200**, and an auxiliary unit **300**. In some examples, sensors of wearable system **601** provide sensor data **608** (which may be multichannel sensor data such as in examples where one or more sensors, including sensors from two or more sensor types, are presented in parallel). Sensor data **608** may be provided concurrently with microphones **604** detecting the source speech; that is, the sensors of wearable system **601** can provide sensor data **608** corresponding to conditions at the time the source speech is provided. In some examples, one or more of microphones **604** can be included in a wearable head device, a handheld controller, and/or an auxiliary unit of wearable system **601** and/or in an environment thereof.

[0038] In example system **600**, subsystem **610** can accept as input sensor data **608** and audio signal **606**; determine control parameters to apply to process audio signal **606**; and provide the processed audio signal as input to a signal processing engine (e.g., speech recognition engine **650** and/or natural language processing (NLP) engine **670**), which may generate an output (e.g., text output **660** and/or natural language processing output **680**, respectively). Subsystem **610** includes one or more processes, stages, and/or modules described below and shown in FIG. 6. Subsystem **610** can be implemented using one or more processors (e.g., CPUs, GPUs, and/or DSPs), a memory, a network, a database, and/or any combination of other suitable components. In some examples, some or all of subsystem **610** can be implemented on wearable system **601**: for example, on one or more of a head device (e.g., wearable head device **100**); a handheld controller (e.g., handheld controller **200**); and/or an auxiliary unit (e.g., auxiliary unit **300**). In some examples, some or all of subsystem **610** can be implemented on a device containing microphone **604** (e.g., a smartphone, or a standalone voice assistant device). In some examples, some or all of subsystem **610** can be implemented on a cloud server or another network-enabled computing device. For instance, a local device could perform latency-sensitive functions of subsystem **610**, such as those related to processing audio signals and/or sensor data; while a cloud server or other network-enabled device could perform functions of subsystem **610** that require large computational or memory resources (e.g., training or applying a complex artificial neural network), and transmit the output to the local device. Other implementations of subsystem **610** will be apparent to those skilled in the art and are within the scope of the disclosure.

[0039] In example system **600**, subsystem **610** includes a sensor data analysis stage **620**. Sensor data analysis stage **620** can process and analyze sensor data **608**, for example in order to determine information about the environment of wearable system **601**. In examples where sensor data **608** includes sensor data from disparate sources (e.g., camera data and GPS data), sensor data analysis stage **620** can combine the sensor data ("sensor fusion") according to techniques known to those in the art (e.g., Kalman filters). In some examples, sensor data analysis stage **620** can incorporate data from other sources in addition to sensor data **608**. For instance, sensor data analysis stage **620** can combine sensor data **608** from a GPS unit with map data and/or

satellite data (e.g., from a memory or database storing such data) in order to determine location information based on the output of the GPS unit. As an example, the GPS unit could output, as part of sensor data 608, GPS coordinates corresponding to the latitude and longitude of wearable system 601. Sensor data analysis stage 620 could use the latitude and longitude with map data to identify in what country, town, street, unit (e.g., commercial unit or residential unit), or room the wearable system 601 is located; or to identify nearby businesses or points of interest. Similarly, architectural data (e.g., from public structural records) can be combined with the sensor data 608 to identify a building in which wearable system 601 is located; or weather data (e.g., from a real-time feed of satellite data) can be combined with the sensor data 608 to identify current weather conditions at that location. Other example applications will be apparent and are within the scope of the disclosure.

[0040] On a smaller scale, sensor data analysis stage 620 can analyze sensor data 608 to generate information relating to objects and geometry in the immediate vicinity of wearable system 601, or relating to a user of wearable system 601. For example, using techniques known in the art, data from a LIDAR sensor or radar unit of wearable system 601 can indicate that wearable system 601 is facing a wall located eight feet away, at an angle θ to a normal of that wall; and image data from a camera of wearable system 601 can identify that wall as likely being made of ceramic tile (an acoustically reflective material). In some examples, an acoustic sensor of wearable system 601 can be used to measure acoustic effects that a surface may have on an acoustic signal (e.g., by comparing a reflected signal from the surface to a source signal transmitted to the surface). In some examples, sensor data analysis stage 620 can use sensor data 608 to determine a position and/or orientation of the user of wearable system 601, for example using an accelerometer, gyroscope, or IMU associated with wearable system 601. In some examples, such as in augmented reality or mixed reality applications, stage 620 can incorporate a map or other representation of the user's current environment; for example, in cases where sensors of wearable system 601 have been used to construct a 3D representation of the geometry of a room, that 3D representation data can be used in conjunction with sensor data 608. Similarly, stage 620 can incorporate information such as the materials of nearby surfaces, and acoustic properties of those surfaces; information relating to other users in the environment (e.g., their locations and orientations, and/or the acoustic characteristics of their voices); and/or information about the user of wearable system 601 (e.g., the user's age group, gender, native language, and/or vocal characteristics).

[0041] In some examples, sensor data analysis stage 620 can analyze sensor data 608 to generate information relating to microphones 604. For example, sensor data 608 may provide a position and/or orientation of one or more microphones 604, such as the position and/or orientation relative to wearable system 601. In examples where wearable system 601 includes one or more microphones 604, the position and orientation of one or more microphones 604 may be directly linked to the position and orientation of the wearable system 601. In some examples, wearable system 601 can include one or more additional microphones that are not one or more microphones 604; such additional one or more microphones can be used to provide a baseline audio signal, for example corresponding to a user's speech as detected from a known

position and orientation, and from a short distance. For example, the additional one or more microphones may be in known positions with known orientations in an environment of the user. The amplitude, phase, and frequency characteristics of this baseline audio signal can be compared to audio signal 606 to identify a relationship between the source speech 602 and audio signal 606. For example, if a first audio signal detected at a first time has an amplitude one-half that of the baseline audio signal, and a second audio signal detected at a second time has an amplitude one-fourth that of the baseline audio signal, it may be inferred that the user has moved away from microphone 604 during the interval between the first time and the second time. This can be extended to any suitable number of microphones (e.g., an initial microphone, and two or more additional microphones).

[0042] Information output by sensor data analysis stage 620, based on sensor data 608 and/or other data as described above, can identify a relationship between source speech 602 and corresponding audio signal 606. Information describing this relationship can be used at stage 630 to compute one or more control parameters that can be applied to audio signal 606, to one or more microphones 604, and/or to wearable system 601. Application of these control parameters can improve the accuracy with which system 600 can recover the underlying source speech 602 from audio signal 606. In some examples, control parameters computed at stage 630 can include digital signal processing (DSP) parameters for that can be applied to process audio signal 606. For instance, such control parameters can include parameters for a digital signal processing (DSP) noise reduction process (e.g., a signal threshold below which gated noise reduction will be applied to audio signal 606, or a noisy frequency of audio signal 606 to be attenuated); parameters for a DSP echo cancellation or dereverberation process (e.g., a time value corresponding to a delay between audio signal 606 and an echo of that signal); or parameters for other audio DSP processes (e.g., phase correction, limiting, pitch correction).

[0043] In some examples, control parameters can define a DSP filter to be applied to audio signal 606. For instance, sensor data 608 (e.g., from a microphone of a head-mounted unit of wearable system 601) can indicate a characteristic frequency curve corresponding to the voice of the user of the wearable system 601 (i.e., the user generating source speech 602). This frequency curve can be used to determine control parameters defining a digital bandpass filter to apply to audio signal 606; this bandpass filter can isolate frequencies corresponding closely to source speech 602, in order to make source speech 602 more prominent in audio signal 606. In some examples, sensor data 608 (e.g., from a microphone of a head-mounted unit of wearable system 601) can indicate a characteristic frequency curve corresponding to the voice of a different user (other than the user of the wearable system 601) in the vicinity of the wearable system 601. This frequency curve can be used to determine control parameters defining a digital notch filter to apply to audio signal 606; this notch filter can remove unwanted voices from audio signal 606, in order to render source speech 602 more prominent in audio signal 606. Similarly, sensor data 608 (e.g., a camera of wearable system 601) can identify specific other individuals in the vicinity, and their positions relative to the wearable system 601; this information can determine a level of the notch filter (e.g., the closer an

individual is, the louder his or her voice is likely to be in audio signal 606, and the greater a level attenuation that may need to be applied). As another example, the presence of certain surfaces and/or materials in the vicinity of a user may affect the frequency characteristics of that user's voice as detected by microphone 604; for instance, if the user is standing in the corner of a room, certain low frequencies of the user's voice may be accentuated in audio signal 606; this information can be used to generate parameters (e.g., a cutoff frequency) of a high-pass filter to be applied to audio signal 606. These control parameters can be applied to audio signal 606 at stage 640; or as part of speech recognition engine 650 and/or natural language processing engine 670 (e.g., at a stage corresponding to signal preprocessing stage 520 or feature extraction stage 530 described above).

[0044] In some examples, control parameters computed at stage 630 can be used to configure microphone 604. For example, such control parameters can include hardware configuration parameters, such as gain levels for a hardware amplifier coupled to microphone 604; beamforming parameters to adjust a directionality of microphone 604 (e.g., a vector at which microphone 602 should be directed); parameters to determine which of a plurality of microphones 604 should be enabled or disabled; or parameters to control where a microphone 604 should be positioned or oriented (e.g., in examples where microphone 604 is attached to a mobile platform). In some examples, microphone 604 may be a component of a smartphone, or another mobile device; and the control parameters computed at stage 630 can be used to control the mobile device (e.g., to enable various components of the mobile device, or to configure or operate software on the mobile device).

[0045] In some examples, control parameters computed at stage 630 can be used to control wearable system 601 itself. For example, such control parameters could include parameters for presenting a message to a user of wearable system 601, such as via displays 110A/110B or speakers 120A/120B (e.g., an audio or video message that the user should move away from a nearby wall in order to improve speech recognition accuracy); or parameters to enable, disable, or reconfigure one or more sensors of wearable system 601 (e.g., to reorient a servo-mounted camera in order to obtain more useful camera data). In examples where the wearable system 601 includes microphones 604, control parameters can be sent to wearable system 601 in order to control microphones 604 such as described above.

[0046] In some examples, control parameters computed at stage 630 can be used to affect a decoding process (e.g., decoding process 540 described above) of a speech processing system (e.g., speech recognition engine 650 and/or natural language processing engine). For example, sensor data 608 may indicate characteristics of a user's environment, behaviors, or mental state that may affect the user's use of language. For instance, sensor data 608 (e.g., from a camera and/or a GPS unit) may indicate that the user is attending a football game; because the user's speech (i.e., source speech 602) may be much more likely than usual to include football-related words (e.g., "coach," "quarterback," "touchdown") while the user is attending a football game, a control parameter of a speech processing system (e.g., language modeling stage 560) can be temporarily set to reflect a higher probability that an audio signal corresponds to a football-related word.

[0047] In some examples, such as shown in FIG. 6, individual update modules 632, 634, and 636 can determine control parameters computed at stage 630, or can apply the control parameters computed at stage 630, and/or sensor data 608, to audio signal 606; microphone 604; wearable system 601; or to any hardware or software subsystem of the above. For example, a beamformer update module 632 can determine, based on sensor data 608 or one or more control parameters computed at stage 630, how a beamforming module (e.g., a beamforming module of microphone 604) can be updated to improve the recognition or natural language processing of audio signal 606. In some examples, beamforming update module 632 can control the directivity of a sensor array (e.g., an array of microphones 604) in order to maximize a signal/noise ratio of a signal detected by the sensor array. For instance, the beamforming update module 632 can adjust the directivity of microphones 604 such that source speech 602 is detected with a minimum of noise and distortion; for example, in a room with multiple voices, software of an adaptive beamforming module can direct microphones 604 to maximize the signal power corresponding to a voice of interest (e.g., the voice corresponding to source speech 602). For instance, sensors of wearable system 601 can output data indicating that wearable system 601 is located a certain distance from microphones 604, from which a time-of-flight value of the source speech 602 to microphones 604 can be determined. This time-of-flight value can be used to calibrate the beamforming, using techniques familiar to those skilled in the art, in order to maximize the ability of speech recognition engine 650 to identify source speech 602 from audio signal 606.

[0048] In some examples, a noise reduction update module 634 can determine, based on sensor data 608 or one or more control parameters computed at stage 630, how a noise reduction process can be updated to improve the recognition or natural language processing of audio signal 606. In some examples, noise reduction update module 634 can control parameters of a noise reduction process applied to audio signal 606 in order to maximize a signal/noise ratio of audio signal 606; this in turn can facilitate automatic speech recognition performed by speech recognition engine 650. For instance, noise reduction update module 634 can selectively apply signal attenuation to frequencies of audio signal 606 on which noise is likely to be present, while boosting (or declining to attenuate) frequencies of audio signal 606 which carry information of the source speech 602. Sensors of wearable system 601 can provide data to help noise reduction update module 634 identify which frequencies of audio signal 606 are likely to correspond to noise, and which frequencies are likely to carry information about the source speech 602. For instance, sensors of wearable system 601 (e.g., GPS, LIDAR, etc.) may identify that wearable system 601 is located on an airplane. An airplane may be associated with background noise having certain characteristic frequencies; for example, airplane engine noise may be concentrated primarily around a known frequency f_0 . Based on this information from the sensors, noise reduction update module 634 can attenuate frequency f_0 of audio signal 606. Similarly, sensors of wearable system 601 (e.g., a microphone mounted to wearable system 601) can identify a frequency signature corresponding to the voice of a user of wearable system 601; noise reduction update module 634 can apply a bandpass filter to a frequency range correspond-

ing to that frequency signature, or can ensure that noise reduction is not applied to that frequency range.

[0049] In some examples, an echo cancellation (or dereverberation) update module **636** can determine, based on sensor data **608** or one or more control parameters computed at stage **630**, how an echo cancellation unit can be updated to improve the recognition or natural language processing of audio signal **606**. In some examples, echo cancellation update module **636** can control parameters of an echo cancellation unit applied to audio signal **606** in order to maximize the ability of a speech recognition engine **650** to determine source speech **602** from audio signal **606**. For instance, echo cancellation update module **636** can instruct an echo cancellation unit to detect and correct for (e.g., via a comb filter) echoes in audio signal **606** that follow the source speech by 100 milliseconds. Because such echoes can interfere with the ability of speech processing systems (e.g., speech recognition engine **650**, natural language processing engine **670**) to determine source speech (e.g., by affecting the ability to extract features, such as described above with respect to FIG. **5** at stage **530**, from audio signal **606**), removing these echoes can result in higher accuracy of speech recognition, natural language processing, and other speech processing tasks. In some examples, sensors of wearable system **601** can provide sensor data **608** that can be used to determine control parameters for echo cancellation. For example, such sensors (e.g., camera, LIDAR, radar, acoustic sensors) can determine that a user of wearable system **601** is located ten feet from a surface, and faces that surface at an angle θ_1 to a normal of that surface, and further that microphone **604** is located twenty feet from that surface, and faces the surface at an angle θ_2 . It can be computed from this sensor data that the surface is likely to produce an echo that reaches microphone **604** at a certain time after the source signal **602** (i.e., the time-of-flight from the user to the surface plus the time-of-flight from the surface to microphone **604**). Similarly, it can be determined from the sensor data that the surface corresponds to a bathroom tile surface, or another surface with known acoustic reflective properties; and those reflective properties can be used to compute control parameters of a dereverberation unit that will attenuate the resultant acoustic reflections in audio signal **606**.

[0050] Similarly, in some examples, signal conditioning can be applied to speech audio signal **606** to account for equalization applied to the speech audio signal **606** by an acoustic environment. For instance, a room can increase or attenuate certain frequencies of speech audio signal **606**, for example by virtue of the room's geometry (e.g., dimensions, cubic volume), materials (e.g., concrete, bathroom tile), or other characteristics that can affect the signal as it is detected by a microphone (e.g., microphone **604**) in the acoustic environment. These effects can complicate the ability of speech recognition engine **650** to perform consistently across different acoustic environments. Sensors of wearable system **601** can provide sensor data **608** which can be used to counteract such effects. For instance, sensor data **608** can indicate, as described above, a size or shape of a room, or the presence of acoustically significant materials, from which one or more filters can be determined, and applied to the audio signal to counteract room effects. In some examples, sensor data **608** can be provided by sensors (e.g., camera, LIDAR, radar, acoustic sensors) that indicate a cubic volume of a room in which the user is present; an acoustic effect of

that room can be modeled as a filter; and an inversion of that filter can be applied to an audio signal to compensate for that acoustic effect.

[0051] In some examples, modules **632**, **634**, or **636**, or other suitable elements of example system **600** (e.g., stage **620** or stage **640**) can make use of predetermined mappings to determine control parameters based on sensor data **608**. In some examples described above, control parameters are computed directly based on sensor data **608**. For instance, as described above, an echo cancellation update module **636** can determine control parameters for an echo cancellation unit that can be applied to attenuate echoes present in audio signal **606**. As described above, such control parameters can be computed by geometrically determining the distance between wearable system **601** and a microphone **604**; computing the time-of-flight that an audio signal traveling at the speed of sound in air would require to travel from the wearable system **601** to the microphone; and setting an echo period of the echo cancellation unit to correspond to that time-of-flight. In some cases, though, the control parameters can be determined by comparing sensor data **608** to a mapping of sensor data to control parameters. Such a mapping could be stored in a database, such as on a cloud server or another networked device. In some examples, one or more elements of example system **601** (e.g., beamformer update module **632**, noise reduction update module **634**, echo cancellation update module **636**, stage **630**, stage **640**, speech recognition engine **650**, and/or natural language processing engine **670**) can query a database to retrieve one or more control parameters that correspond to sensor data **608**. This process can occur instead of, or in addition to, direct computation of control parameters such as described above. In such a process, sensor data **608** could be provided to the predetermined mapping; and one or more control parameters that most closely correspond to sensor data **608** in the predetermined mapping can be returned. Using a predetermined mapping of sensor data to parameters can carry several advantages. For instance, performing a lookup from a predetermined mapping can be computationally less expensive than processing sensor data in real-time-particularly when the computations may involve complex geometric data, or where sensors of wearable system **601** may suffer from significant latency or bandwidth limitations. Further, predetermined mappings can capture relationships between sensor data and control parameters that may be difficult to compute from sensor data alone (e.g. by mathematical modeling).

[0052] In some examples, machine learning techniques can be used to generate or refine mappings of sensor data to control parameters, or to otherwise determine control parameters from sensor data **608**. For example, a neural network (or other appropriate machine learning technology) can be trained to identify desired control parameters based on sensor data input, according to techniques familiar to those skilled in the art. Desired control parameters can be identified, and the neural network further refined, through user feedback. For example, users of system **600** can be prompted to rate the quality of speech recognition output (e.g., text output **660**) and/or natural language processing output (e.g., natural language processing output **680**). Such user ratings can be used to adjust the likelihood that a given set of sensor data will result in a particular set of control parameters. For example, if users report a high rating of text output **660** for a particular set of control parameters and a

particular set of sensor data, a mapping of that sensor data to those control parameters can be created (or a link between them strengthened). Conversely, a low rating can cause a mapping of the sensor data to the control parameters to be weakened or deleted.

[0053] Similarly, machine learning techniques can be utilized to improve the ability of a speech recognition engine (e.g., **650**) to discriminate between speech belonging to a user, and speech belonging to other entities—for example, speech emanating from a television or a stereo system. As described above, neural network (or other appropriate machine learning technology) can be trained to perform this discrimination (i.e., determining whether input audio belongs to the user, or to some other source). In some cases, a calibration routine, in which the user provides a set of input audio known to belong to that user, can be used to train the neural network. Other suitable machine learning technologies can be used for the same purpose.

[0054] While update modules **632**, **634**, and **636** are described to comprise a beamforming update module, a noise reduction update module, and an echo cancellation module, respectively, other suitable modules may be included in any combination. For instance, in some examples, an EQ update module can determine how filtering processes (such as described above) can be updated, based on sensor data **608** and/or one or more control parameters computed at stage **630**. Moreover, the functions described above with respect to modules **632**, **634**, and **636** can be performed as part of other elements of example system **600**, such as stage **630** and/or stage **640**, or speech recognition engine **650** or natural language processing engine **670**.

[0055] Although the disclosed examples have been fully described with reference to the accompanying drawings, it is to be noted that various changes and modifications will become apparent to those skilled in the art. For example, elements of one or more implementations may be combined, deleted, modified, or supplemented to form further implementations. Such changes and modifications are to be understood as being included within the scope of the disclosed examples as defined by the appended claims.

1. A method of processing an acoustic signal, the method comprising:

- receiving the acoustic signal, the acoustic signal associated with a user of a wearable device;
- receiving, via one or more sensors, one or more sensor inputs, wherein the one or more sensor inputs indicate a material associated with a location of the user;
- identifying, based on the one or more sensor inputs, the material associated with the location;
- determining a cutoff frequency of a filter based on the material associated with the location; and
- applying the filter to the acoustic signal.

2. The method of claim 1, wherein the filter is applied to the acoustic signal to generate a second acoustic signal, and the method further comprises providing the second acoustic signal to a speech recognition engine to generate text output associated with the acoustic signal.

3. The method of claim 1, wherein the filter is applied to the acoustic signal to generate a second acoustic signal, and the method further comprises providing the second acoustic signal to a natural language processing engine to generate natural language data associated with the acoustic signal.

4. The method of claim 1, further comprising:
determining control parameters based on the one or more sensor inputs; and

applying the control parameters to the acoustic signal.

5. The method of claim 4, further comprising:

detecting, based on the one or more sensor inputs, a surface; and

determining an effect of the surface on the acoustic signal, wherein the control parameters cause a reduction of the effect of the surface on the acoustic signal.

6. The method of claim 5, further comprising determining an acoustic property of the surface, wherein the effect of the surface on the acoustic signal is determined based on the acoustic property.

7. The method of claim 4, further comprising:

detecting, based on the one or more sensor inputs, a person different from the user; and

determining an effect of the person on the acoustic signal, wherein the control parameters cause a reduction of the effect of the person on the acoustic signal.

8. The method of claim 4, wherein the control parameters comprise a control parameter for echo cancellation, and said determining the control parameters based on the one or more sensor inputs comprises:

detecting, based on the one or more sensor inputs, a surface; and

determining a time of flight between the surface and a microphone for receiving the acoustic signal.

9. The method of claim 4, wherein the control parameters comprise a control parameter for beamforming, and said determining the control parameters based on the one or more sensor inputs comprises determining a time of flight between the user and a microphone for receiving the acoustic signal.

10. The method of claim 4, further comprising determining a three-dimensional representation of an environment of the wearable device; and wherein said determining the control parameters is further based on the three-dimensional representation of the environment of the wearable device.

11. The method of claim 1, further comprising:

determining a quality of the acoustic signal;

in accordance with a determination that the quality of the acoustic signal is below a quality threshold, presenting a message to the user to move from the location to a new location to improve the quality of the acoustic signal.

12. The method of claim 1, wherein the wearable device comprises the one or more sensors.

13. The method of claim 1, wherein the filter comprises a high-pass filter.

14. The method of claim 1, further comprising determining, based on the one or more sensor inputs, the location of the user.

15. The method of claim 1, wherein the acoustic signal is received via a microphone.

16. A system comprising:

a wearable device;

one or more sensors; and

one or more processors configured to perform a method comprising:

receiving an acoustic signal, the acoustic signal associated with a user of the wearable device;

receiving, via the one or more sensors, one or more sensor inputs, wherein the one or more sensor inputs indicate a material associated with a location of the user;

identifying, based on the one or more sensor inputs, the material associated with the location;

determining a cutoff frequency of a filter based on the material associated with the location; and

applying the filter to the acoustic signal.

17. The system of claim **16**, the method further comprising:

determining control parameters based on the one or more sensor inputs; and

applying the control parameters to the acoustic signal.

18. The system of claim **16**, wherein the wearable device comprises the one or more sensors.

19. The system of claim **16**, wherein the acoustic signal is received via a microphone.

20. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to perform a method comprising:

receiving an acoustic signal, the acoustic signal associated with a user of a wearable device;

receiving, via one or more sensors, one or more sensor inputs, wherein the one or more sensor inputs indicate a material associated with a location of the user;

identifying, based on the one or more sensor inputs, the material associated with the location;

determining a cutoff frequency of a filter based on the material associated with the location; and

applying the filter to the acoustic signal.

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